Q1. Power systems optimisation & algorithms

(a) Give an overview of the three main types of power allocation problems in power systems. Note their uses in industry, advantages and disadvantages.

(b) Explain what a "penalty factor" in economic dispatch is.

(c) What is a security constant in an optimal power flow problem?

(d) An island power company with a network comprising 10 controllable generators, 100 buses and 100 branches wants to run a security-constrained AC optimal power flow. If there are five contingencies which must be fully included, estimate the size of the problem. The size of the problem is defined by the number of design variables and number of constraints. There are no other design variables except those associated with the generators (i.e. any regulators, tap settings, etc. are not included in the OPF problem).

(e) For the same scenario as the previous question, estimate the size of the economic dispatch, DC OPF, and AC-OPF (no security constraints) problems. In addition to scaling / size issues, what makes the the AC-OPF and SC-ACOPF difficult to solve?

(f) Consider the three-bus system depicted in Fig. 1. You may assume that each meter is very close to the nearest bus in the figure.

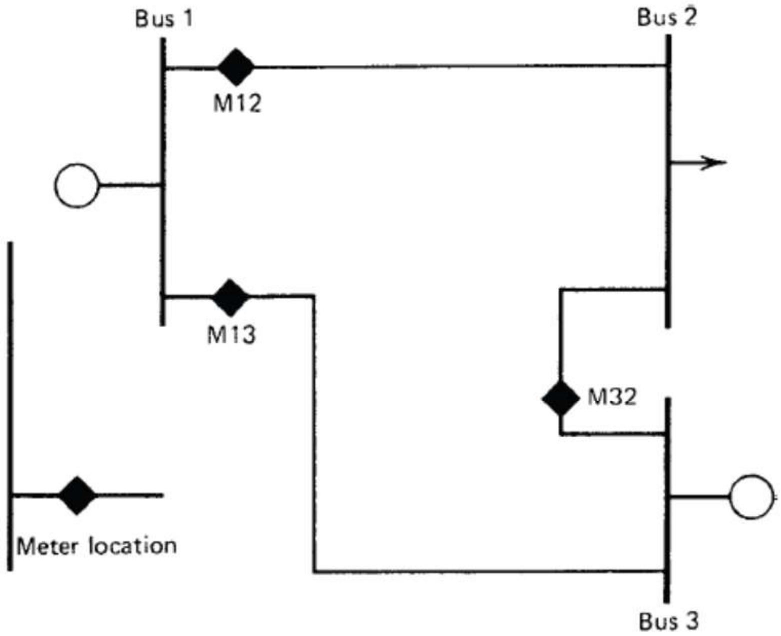

Figure 1: Power system for question 4 (e)

a) Using the DC power-flow approximation derived in lectures, solve for the angles of the network if bus 1 (not bus 3 as in the lecture notes) is taken as the reference. The network impedances are $X_{12} = 0.2, X_{13} = 0.4, X_{23} = 0.25$ and the measurements were M_{12} = 62 MW, M_{13} = 6 MW, M_{32} = 37 MW.

b) Taking bus 3 as a reference (as in the lecture example), suppose the measurement M_{13} is taken by a better meter with $\pm$ 0.3 MW accuracy. With the aid of a computer if necessary, calculate the weighted least squares estimate of the voltage angles and buses 1 and 2.

c) If the DC power-flow approximation is **NOT** used, and **all bus voltages, along with reactive power at the meter locations, are measured** as well, formulate (do NOT solve) an appropriate non-linear least squares problem. Assume all measurements (including those of the bus voltages) have the same standard deviation in p.u. and thus weightings can be ignored.

The active power flow through the line from i to j is given by:
$P_{ij} = V_i V_j B_{ij} \sin(\theta_i - \theta_j)$.

The reactive power leaving bus i through the line from i to j is given by:
$Q_{ij} = V_i(V_i - V_j \cos(\theta_i - \theta_j))B_{ij}$.

Extra Answer Space (Optional)